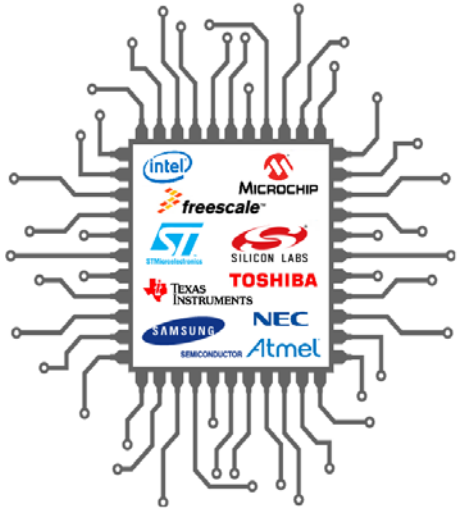
Introduction to Microcontrollers



Thareendhra Keerthi Wijayasiriwardhane

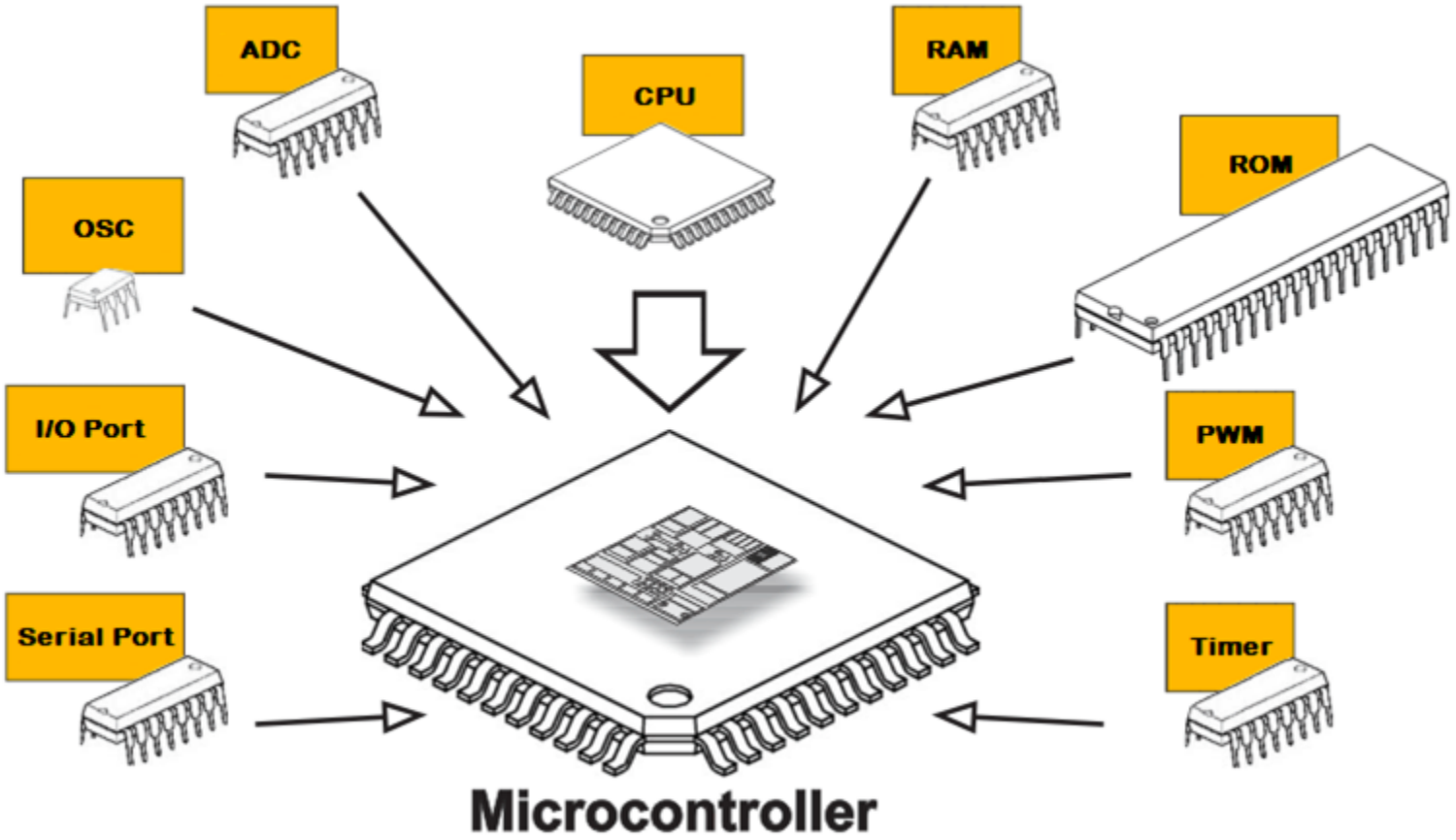
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Learning Outcomes

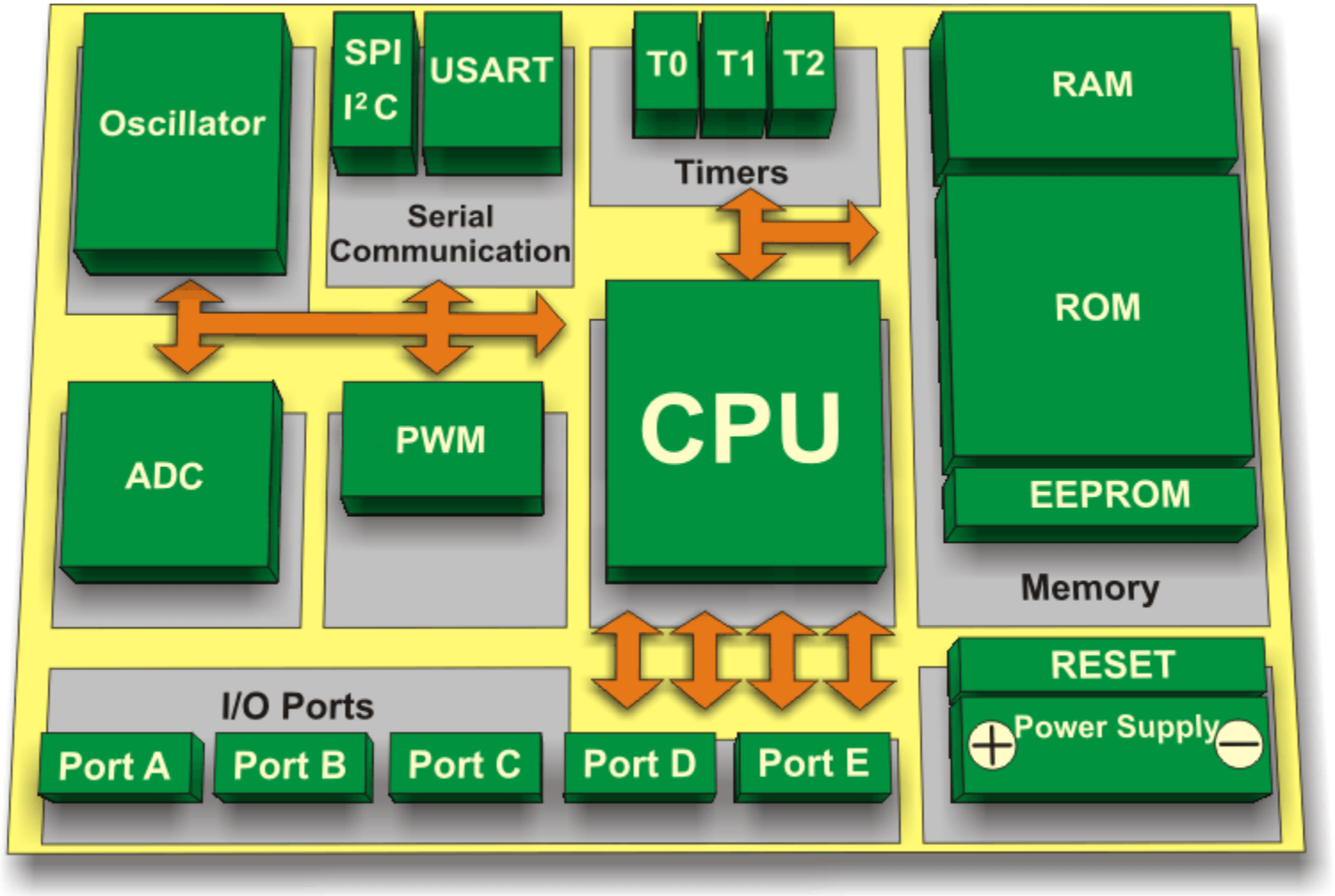
- On completion of this session, you should be able to
  - Explain the functionality of the core components of a microcontroller
  - Explain the internal working of a microcontroller
  - Explain how microcontrollers are programmed
  - Select a microcontroller suitable for academic purposes
  - Discuss about the AVR family of microcontrollers
  - Explain the main features of ATmega328P
  - Select a suitable microcontroller-based development board for academic purposes

Microcontroller

- What is a microcontroller?

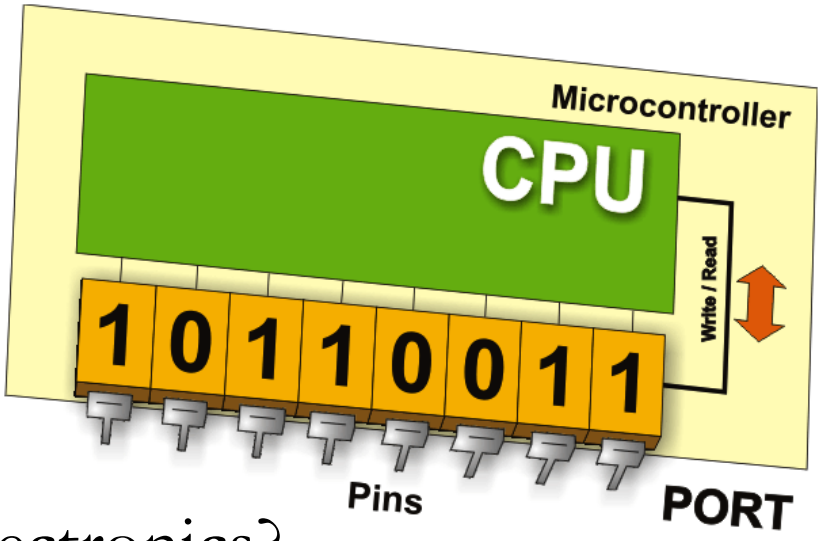


Core Components

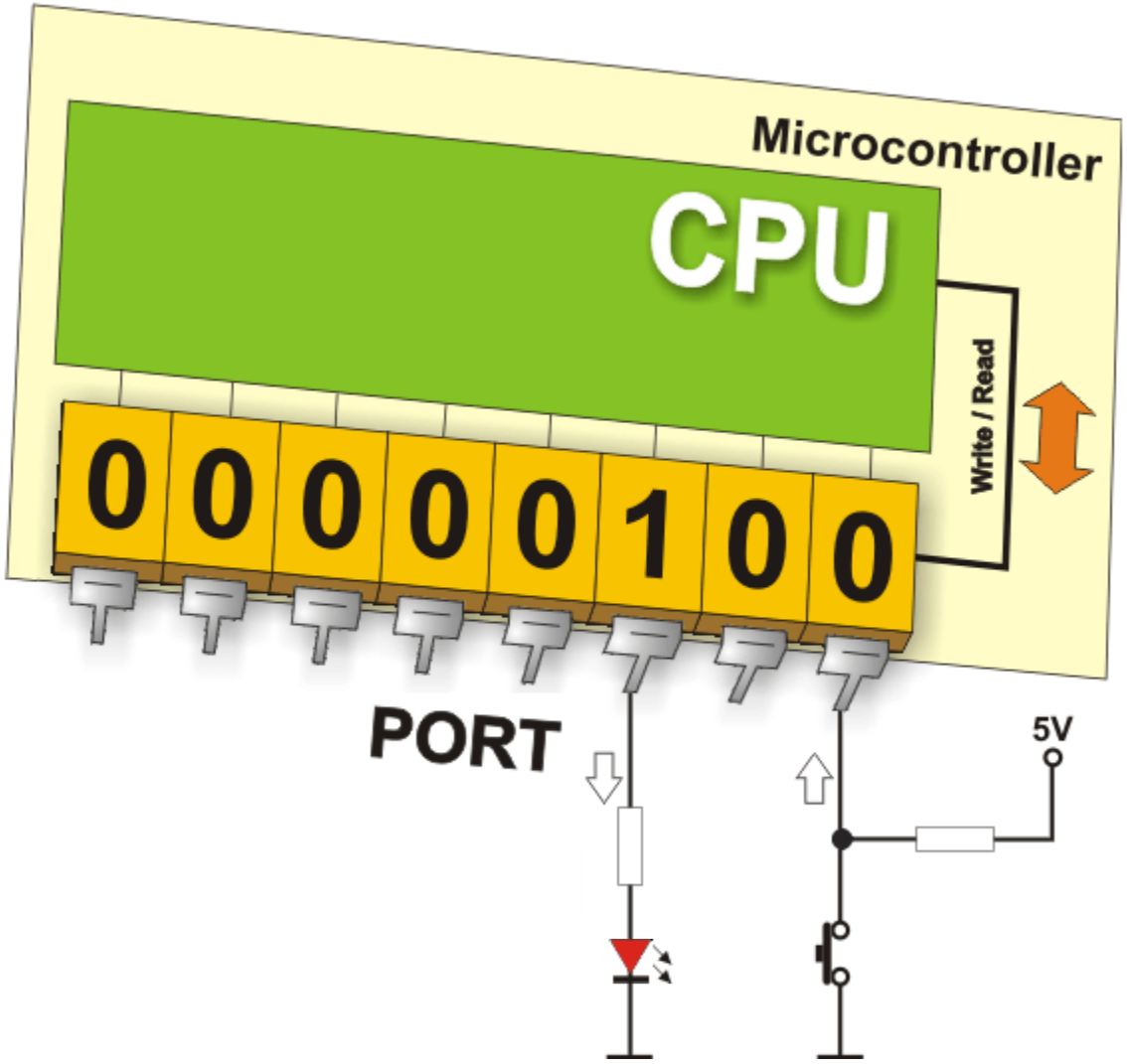


I/O Port

- What's an I/O Port?
- How can external electronics alter the flow of program execution?
- How can program in execution drive external electronics?

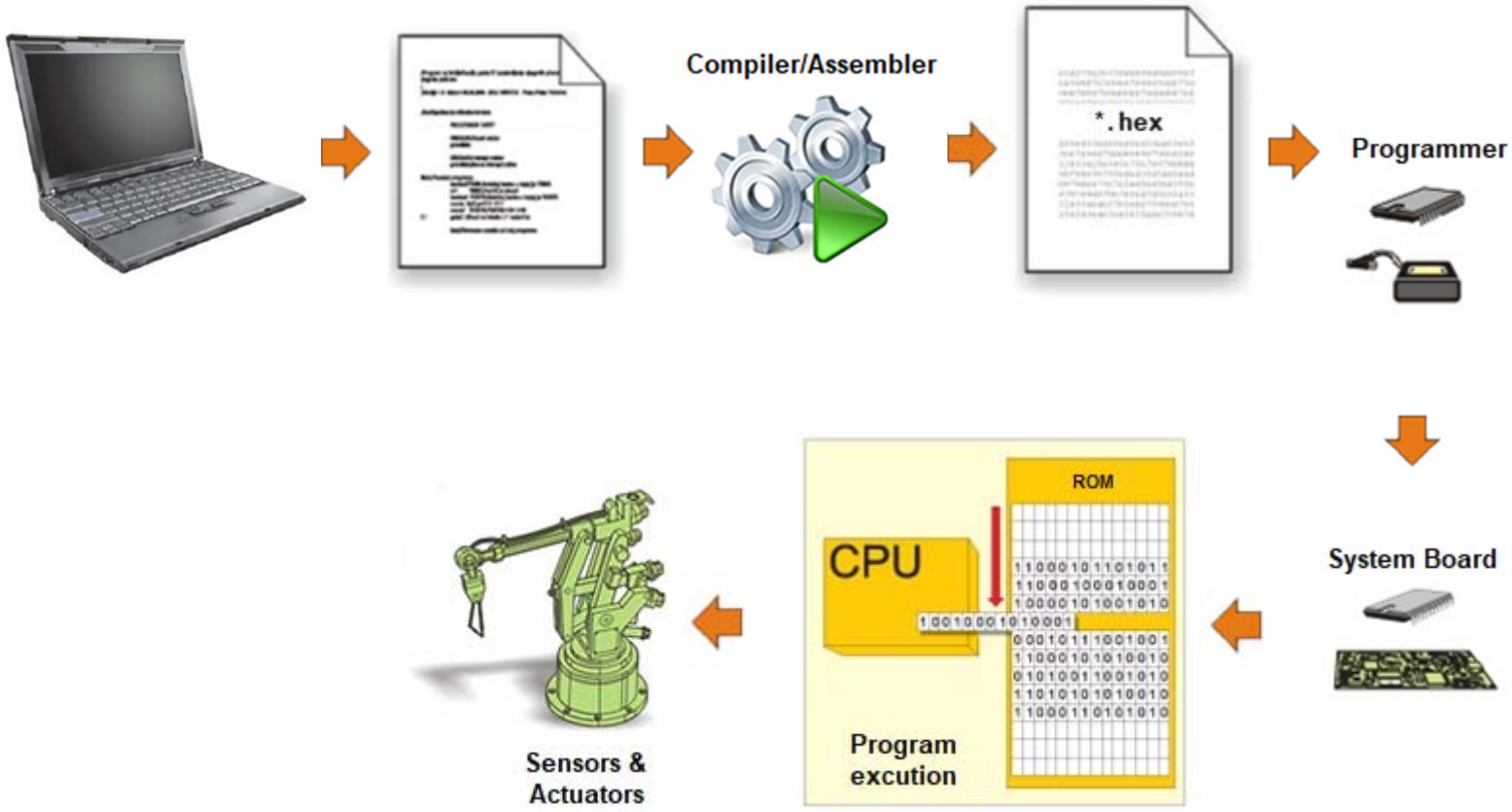


Internal Working



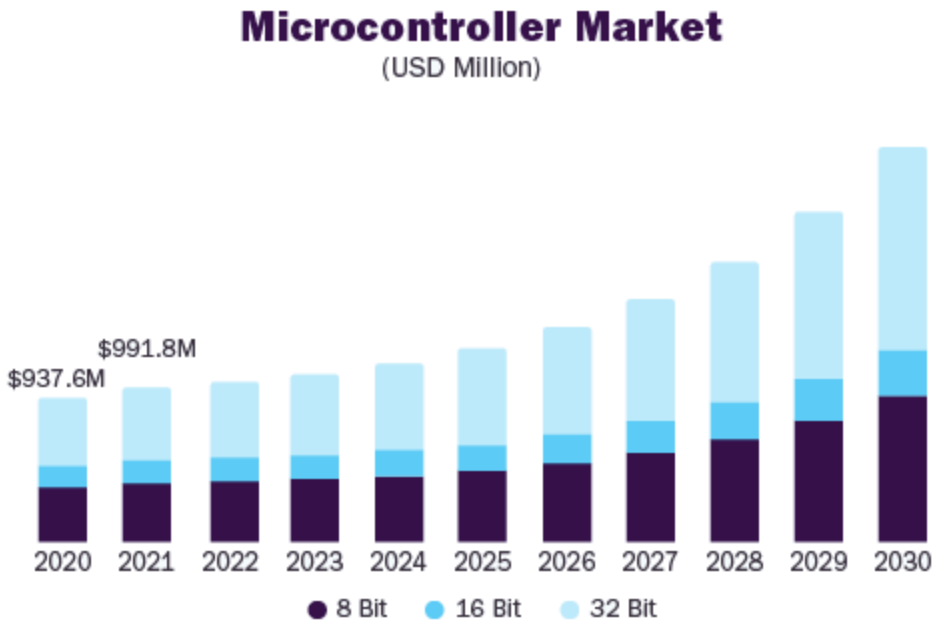

```
Pseudocode
if bit 0 of PORT = 0
    bit 2 of PORT ← 1
else
    bit 2 of PORT ← 0
end if
```

Programming



Word Size

- 8, 16 or 32-bit?
  - Trend





  - Ease of learning
  - Cost

Families

- 8-bit families
  - 68HC08/68HC11 from NXP (formerly Motorola)
  - 8051 from Intel
  - AVR from Microchip (formerly Atmel)
  - PIC from Microchip
  - STM8 from STMicroelectronics
- Why AVR ?

AVR Microcontrollers



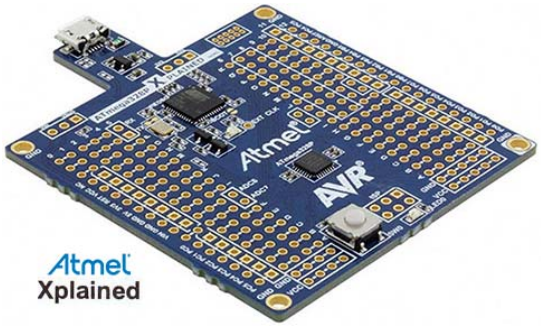
ATmega328P



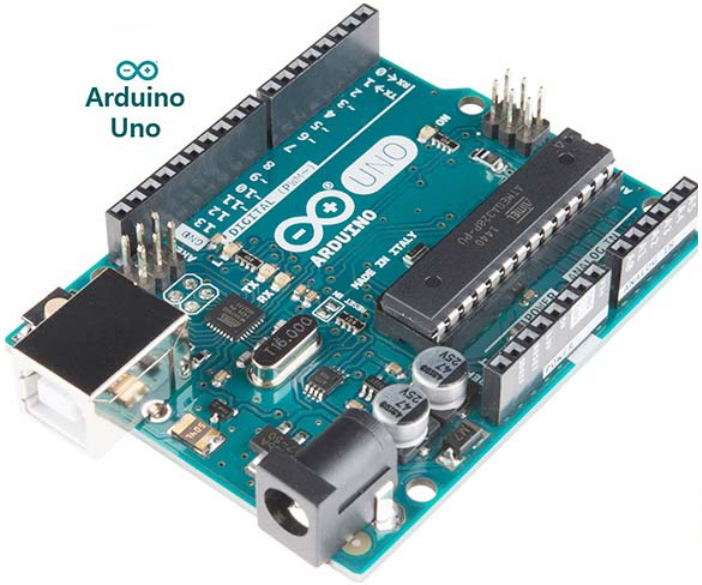
- Features
  - Harvard RISC Architecture
  - 131 single clock cycle instructions (mostly)
  - 8-bit 20 MHz CPU with 14-bit PC
  - 32 KB 16-bit word addressable program ROM
  - 2 KB byte addressable RAM and 1 KB EEPROM
  - Three I/O ports with 23 I/O pins
  - 3 Timers, 6 PWM channels and 6-channel 10-bit ADC
  - One USART, two SPI and an I^2C
  - 28 pin PDIP
- Why ATmega328P?

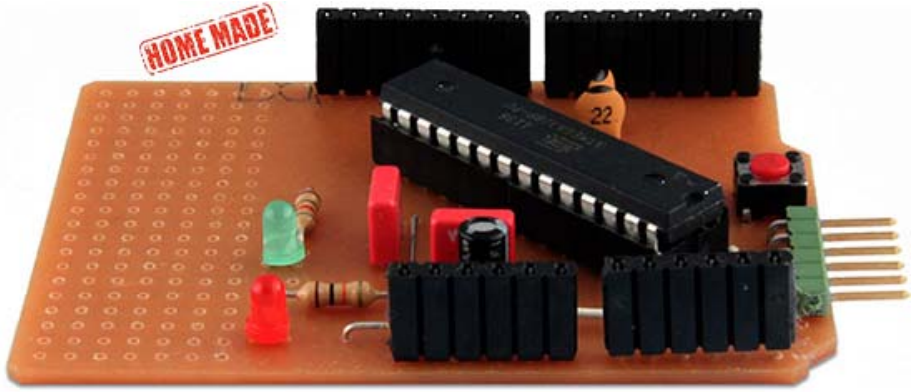
Development Boards

- Microcontroller-based









Arduino

- An open-source, low cost, easy-to-use hardware and software platform for prototyping embedded systems


Uno


Mega


101


Zero


Yun


Lilypad

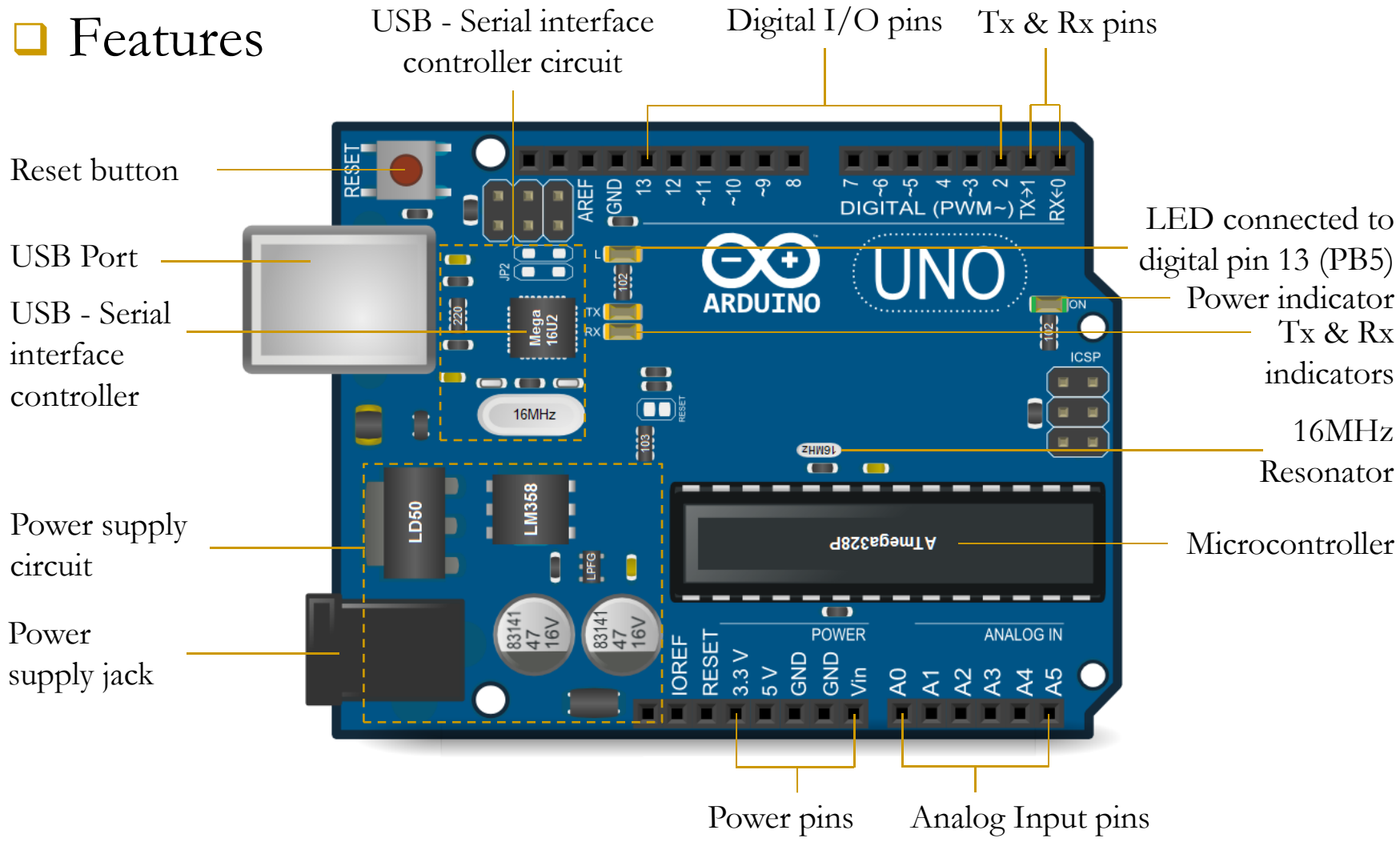

Nano


MKR Zero

- For more information, visit https://www.arduino.cc/

Arduino Uno

- Features



Questions